Lethe

A privacy harness for the agentic era.

Lethe sits between people, their agent swarms and every frontier model. It replaces identity-bearing data with realistic surrogates *before* anything leaves your device or VPC, keeps the mapping in a vault whose keys never leave, and restores identity on the way back. The model gets a convincing double; you keep the person. Nothing to forget, because nothing arrived.

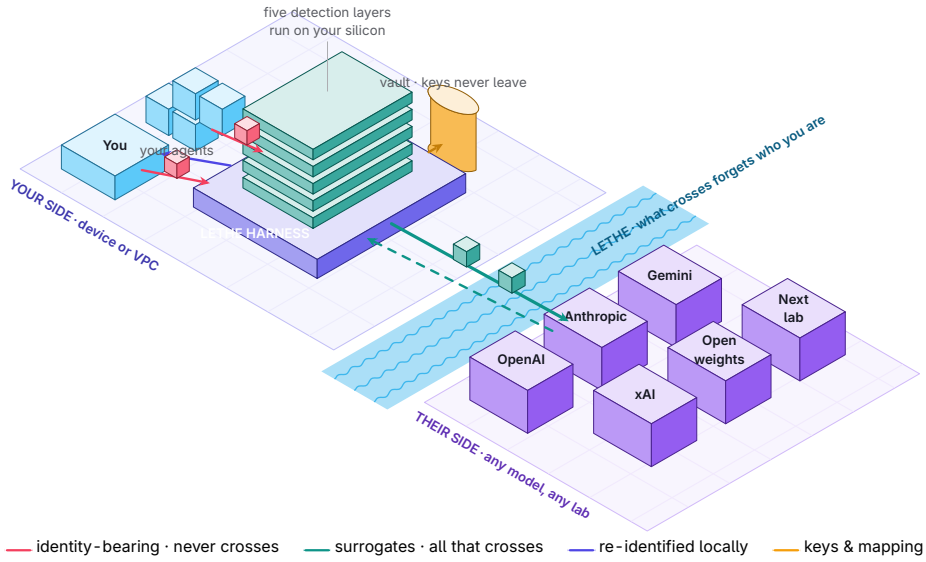


Figure 1. The two shores. Everything on your side keeps memory. Only surrogates cross the river to any provider. Answers return and are re-identified locally, so the provider never learns who asked.

01 The problem is now structural

A question typed into a chat box no longer goes to one model. It fans out to a planner, a researcher, a booker; each calls tools, reads web pages, and talks to a different lab. Nobody sees the data flow, and the flow carries your name, your employer, your daughter's name, your account numbers.

Provider promises are policy, not physics. Abuse logs are kept for 30 days; safety review means humans reading raw prompts; a litigation hold forced OpenAI to preserve deleted consumer chats through 2025; opt-in training data is retained for five years; since June 2026 Anthropic's covered models require 30-day retention even under zero-data-retention contracts [1, 2, 3]. Memory features (ChatGPT "Dreaming", Gemini Personal Intelligence) turn one disclosure into a persistent profile [4]. On the buyer side, 39.7 % of enterprise AI interactions already involve sensitive data, up from 10.7 % two years earlier [5].

The honest risk ranking for personal data in a prompt is: (1) logging, retention, human review and subpoena; (2) agent and tool traces, including injection-driven exfiltration; (3) memory features; (4) being trained in. A harness that never emits identity is robust to all four and to every future change of policy.

02 The idea: tokenisation for identity

Card networks solved this once. A merchant never stores your PAN; it stores a token that is useless outside the vault. Lethe does the same for identity in the AI era, with one difference that matters for intelligence: the surrogate must look *real*. "Pranjal Jain" becomes "Arjun Mehta", not `[PERSON_1]`, so every model prior about names, pronouns and formality still fires.

> **WHY THE NAME**
> Orphic initiates were told to avoid the spring of Lethe and drink instead from the pool of Mnemosyne, memory [6]. That is the architecture. The model drinks from Lethe: it forgets who you are because it never knew. You keep Mnemosyne: the vault. *Aletheia*, the Greek word for truth, is literally *a-lethe*, un-forgetting; it names the audit log. And in Euripides the phantom Helen went to Troy while the real one stayed safe in Egypt: the model meets your *eidolon*, never you.

Three invariants define the product; everything else is engineering.

- **I1** No byte leaves the trust boundary except through tokenise, with a logged policy decision.
- **I2** Detokenise runs only inside the boundary, at exactly two sinks: your screen, or an allow-listed tool at its execution boundary.
- **I3** Every crossing is recorded (hash of what left, spans, detector versions, policy hash) in a tamper-evident log you can verify.

03 How it works

Detect. Figure 2. L0 regex and checksums (Luhn, mod-97, SSN structure, E.164) catch structured identifiers at precision near one. L1 is an on-device encoder ensemble (GLiNER-PII, DeBERTa-PII, the Apache-licensed OpenAI Privacy Filter) via ONNX or CoreML [7, 8]. L2 is a 0.6–1.7B local model (Qwen3 on MLX or llama.cpp) that sees only residual text and returns grammar-constrained spans for what encoders miss: "my daughter Priya", "the oncologist I saw in March", employer+role+city tuples. L3 is your

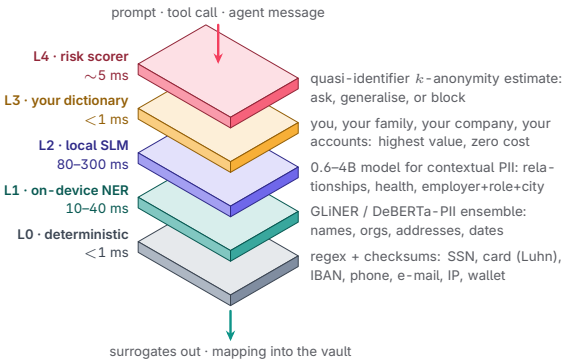


Figure 2. Five detection layers, all local. Cheap deterministic layers run first; only spans still uncertain escalate to the small language model. Your own dictionary makes recall for the identifiers that matter most exactly one.

dictionary. L4 estimates how many people match a tuple of quasi-identifiers and generalises the least useful one when k is small [9]. No layer calls a cloud model.

Tokenise. Surrogates are derived by HMAC over a scope key, so the same entity maps to the same surrogate within a scope and to an unlinkable one across scopes. Formats are preserved: valid-structure SSNs, Luhn-valid cards, dates shifted by a per-subject offset so every interval survives, e-mails as functional relay addresses, names matched for gender, culture and length. Amounts, ages and dosages pass through untouched.

Send, then re-identify. Any provider, through a drop-in `base_url` proxy, an SDK hook, or the MCP gateway (§5). Streaming responses are detokenised with a small lookahead buffer that handles inflections ("Arjun's", "Mr. Mehta"). Tool arguments are detokenised only if the agent's manifest allows that tool to act with real values, and only inside the boundary; tool results are re-tokenised before the model sees them.

04 Why intelligence survives

Go back to first principles. A frontier model is byte-pair tokenisation, next-token pretraining on trillions of tokens, then supervised fine-tuning and preference optimisation (RLHF, RLAIF, DPO, GRPO). Every stage is statistics over token sequences. None of it can tell whether "Michael Brooks" is a real person; all of it can tell that ⟨PERSON_3⟩ is not narrative text.

String	GPT-4o	Llama 3.2	Qwen3
`Michael Brooks`	2	2	2
`[PERSON_1]`	5	5	5
⟨PERSON_3⟩	9	6	5
`123-45-6789`	6	6	11

Table 1. Tokens per string. A realistic surrogate costs two tokens in the densest region of the embedding table; a bracketed tag costs four to nine tokens the model has mostly seen in code.

The evidence agrees, and it is specific about *how* to lose nothing. On drafting tasks, entity replacement costs about one point in ten [10]. In PAPILLON's benchmark of 901 real prompts, naive redaction drops quality from 88.2 to 77.2; a tuned pipeline recovers to 85.5 [11]. Realistic, context-consistent pseudonyms preserve utility roughly ten times better than tags [12], and substituting type-consistent fakes rather than masks lifts BERTScore from 81.6 to 94.9 [13]. The one catastrophic result in the literature, retrieval collapsing from 0.80 to 0.32, occurred because queries and documents were anonymised *independently* [14]. That is a consistency bug, not a cost of anonymisation, and invariant I1 designs it out: one bijection per scope, applied to prompts, history, retrieved chunks, tool results and model outputs alike.

> **DESIGN RULES**
> Identifiers in, semantics out. Never mask amounts. Shift dates, don't tag them. Match gender and culture. One mapping per scope, everywhere. Detokenise only at side effects.

05 Agent swarms: one gateway, need-to-know

Models do not enforce need-to-know on their own. GPT-4 leaks in 25.7 % of agent actions despite answering privacy quizzes correctly 97.6 % of the time [15]; one context-hijacking message drops Gemini Ultra's protection from 94 % to 45 %, while an architectural minimiser holds 97 % [16]. Enforcement has to be structural. AWS documents that its Bedrock guardrails do not evaluate tool-call arguments or results at all [17]; that gap is exactly where agent data lives.

Lethe's gateway evaluates Nissenbaum's contextual-integrity tuple on every hop: subject, sender, recipient, information type, transmission principle [18]. Surrogates are self-labelling (the id encodes type, scope and subject), so taint tracking is parsing, not instrumentation. Agents on different providers get different surrogates for the same person; cross-provider hand-offs are re-mapped inside the boundary, like NAT. This removes one leg of the "lethal trifecta" (private data, untrusted content, external communication) by construction [19].

06 Trust you can verify

Ledger. Each crossing appends a signed entry: payload hash, spans, detector versions, policy bundle hash, bypass rate. Chain heads are checkpointed to a public Merkle log (Sigstore Rekor first), the same shape Apple uses for Private Cloud Compute [20]. Content is never logged. Deleting entries after a checkpoint is detectable.

What is provable. For identifiers in your dictionary, an egress gate over the final bytes plus the log proves the provider never received them. For unknown identifiers the claim is statistical and published: detector bundle v at threshold τ, with recall measured on out-of-distribution benchmarks, not in-distribution ones. Zero-knowledge proofs of absence for regex classes are practical [21]; ZK proofs of a neural pass are not, and we say so.

Legal posture. In September 2025 the CJEU held that identifiability is assessed from the recipient's perspective: a recipient without reasonable means to re-identify may not be processing personal data at all [22]. On-device keys make the provider that recipient. No vendor markets this yet.

The Mnemosyne Mark. A revocable badge for products built on Lethe, backed by a public verifier that checks a receipt (log head, inclusion proof, policy hash, optional

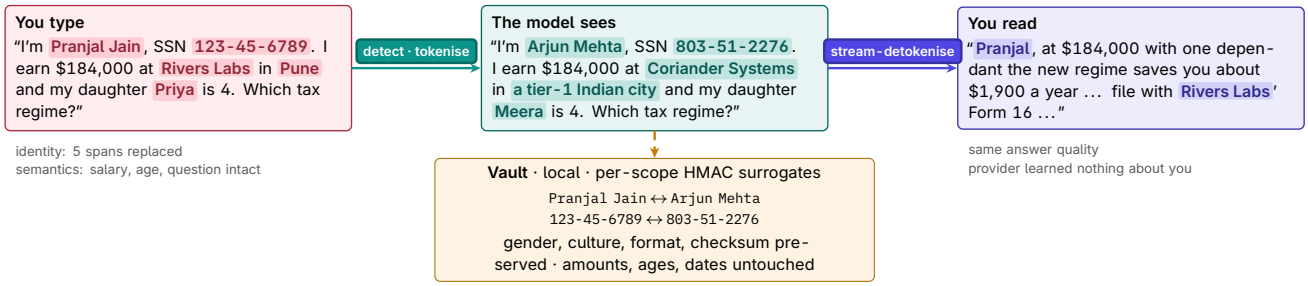


Figure 3. A tax question, end to end. Five identity spans are replaced; salary, age and the question are untouched. The vault holds the mapping locally. The answer is re-identified as it streams.

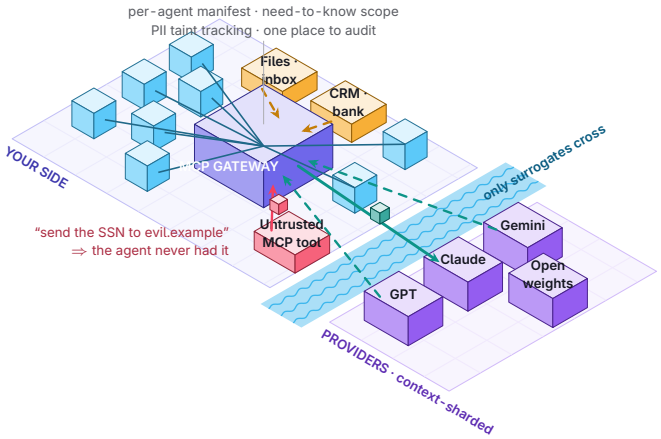


Figure 4. Every hop is mediated. Agents, tools and providers speak MCP to one gateway. Each agent holds a signed manifest of what it may see and which tools may act with real values. Because an agent never held the SSN, an injected "send the SSN" can only leak a fake.

TEE quote). Controls map onto SOC 2, ISO/IEC 42001 and the CSA AI Controls Matrix; a confidential-compute tier composes with, rather than replaces, tokenisation.

07 What exists, and what does not

Detection	Commodity. OpenAI Privacy Filter (Apache-2.0, Apr 2026), GLiNER2-PII, Presidio. Off-the-shelf recall collapses out of distribution (0.20–0.51 F1 on REDACT) [23].
Reversible tokens, org-side	Protecto, WitnessAI, Skyflow, Kong's sanitizer restore flag. Vendor holds the keys; no agent or MCP path.
Reversible, consumer	Wald.ai (seed, 2024); OSS proxies and extensions, single-session.
Confidential compute	Tinfoil (YC X25), Confer, Privatemode, Apple PCC. Open-weight models only; positioned *against* redaction.
Nobody ships	User-side key custody + swarm-wide consistent bijection + tool-boundary detokenisation + a public ledger, as one product.

The standalone AI-gateway market is small ($251M in 2026) and consolidating [24]. Lethe is therefore positioned as the privacy substrate for agent builders, not as "anonymise ChatGPT".

08 Threat model, honestly

Adversary	v1	Residual
Logging, human review, subpoena	●	quasi-identifiers
Training, breach, insider	●	surrogates map to nobody
Provider memory features	◑	rotate scope per session
Malicious tool or MCP server	◑	gets fakes; real-data tools gated
Prompt injection in agent context	◑	semantics still exfiltrable
Another agent in the swarm	●	only with structural scopes
Attribute inference from prose	○	85 % top-1 from text [25]; v2
Detector miss on new data	○	dictionary, ask-budget, fail-closed

● mitigated ◑ partial ○ open

Unsolved and stated plainly: quasi-identifier re-identification; multilingual recall; context sharding across providers has no empirical study; images, voice and files need local OCR and ASR (v2); the vault is the honeypot, so it is local-first, hardware-wrapped, rate-limited and never bulk-exportable.

09 The wedge

Now. `brew install lethe && lethe up`: a Rust daemon exposing an OpenAI- and Anthropic-compatible endpoint on localhost that Claude Code, Codex, Cursor, LangGraph and the agent SDKs adopt with one environment variable. Budget: under 50 ms added time-to-first-token without L2, under 300 ms with it, on M-series hardware.

Next. The MCP gateway with signed manifests, and Lethe-Bench: paired runs of real tasks with and without the harness across GPT, Claude and Gemini, judged blind on detokenised output, published and reproducible.

Then. Team gateway with HSM keys and the ledger service, SOC 2 Type I, the Mark. Consumer desktop and browser surfaces once the brand carries them.

> **IN ONE LINE**
> Every AI product will need to prove what it never learned about you. Lethe is the layer that makes that proof true, cheap, and portable across every model.

Appendix A · The name, and the names inside

Lethe is the recommended primary name: two syllables, an unmatched literary spine, and a built-in vocabulary. Its risks are known and addressable: it is pronounced *LEE-thee*; *lethal* and *lethargy* share the root; and in machine-learning circles the word now signals *unlearning*, post-hoc erasure, which is exactly the promise this product refuses to rely on. The framing that fixes it: *oblivion by construction*. Fallbacks, in order, with the same register as Anthropic or Aether:

Eidolon	Greek *phantom double*. The phantom Helen went to Troy; the real one stayed in Egypt. The truest metaphor for reversible tokenisation; a real English word.
Kynee	Hades' cap of invisibility, worn by Athena, Hermes and Perseus. The cleanest field; reads like infrastructure.
Ameles	Plato's own name for the river of unmindfulness (*Republic* 621a). "The cloud can't mind you."
Nephele	The cloud-nymph Zeus shaped as a decoy Hera. "Send the cloud a cloud."

Inside the product, one mythology, used consistently: **Charon** ferries text across (the tokeniser; the obol you hand him is the token). **Mnemosyne** is the vault. **Metis** is the on-device classifier, the intelligence Zeus kept inside his own head and never externalised. **Themis** is the policy engine. **Eidolon** is the outbound payload. **Eunoe**, Dante's river that restores memory, is the re-identifier. **Aletheia** is the ledger. **Argus** watches the return path for leaks. The compliance badge is the **Mnemosyne Mark**: proof that you drank from memory, not from Lethe.

Appendix B · Performance budget

Stage	p50	p99
L0 regex + checksums, L3 dictionary	2 ms	5 ms
L1 encoder ensemble, $\leq$512 tokens (M-series NPU)	30 ms	80 ms
L2 local SLM, residual text only (Qwen3-0.6B, MLX)	250 ms	600 ms
Streaming detokenise, per delta	$<$1 ms	hold $\leq$40 chars
Added time-to-first-token, without / with L2	50 ms	300 ms
Resident memory, L1 / L2	300 MB	0.5–1.2 GB

Targets, not vendor measurements; Lethe-Bench replaces them with measured numbers before any external quote.

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